

ex $a^0 = 1$

⇒ Define the determinants of a square Matrix

ex: $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$,

$$|A| = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

⇒

$$|A| = a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} +$$

$$a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

ex: $\begin{vmatrix} 3 & 7 \\ 1 & -4 \end{vmatrix} \quad \begin{vmatrix} a & b \\ c & d \end{vmatrix}$

ex: if $A = \begin{vmatrix} 4 & -3 & 5 \\ 1 & 0 & 3 \\ -1 & 5 & 2 \end{vmatrix}$

① $\Rightarrow$ Evaluate each of the following determinants:-

$$\text{i) } \begin{vmatrix} -5 & -3 \\ 3 & -4 \end{vmatrix}$$

sol

$$|A| = \begin{vmatrix} -5 & -3 \\ 3 & -4 \end{vmatrix}$$

ad-bc

$$|A| = \cancel{3} \cdot (-5)(-4) - (3)(\cancel{-3})$$

$$|A| = 20 + 9$$

$$|A| = \cancel{22} \quad 29 \rightarrow \text{Anse}$$

$$\text{ii) } \begin{vmatrix} 3 & -3 & -4 \\ 4 & 1 & -5 \\ 0 & -1 & -4 \end{vmatrix}$$

$$3 \begin{vmatrix} 1 & -5 \\ -1 & -4 \end{vmatrix} - (-3) \begin{vmatrix} 4 & -5 \\ 0 & -4 \end{vmatrix} + (-4) \begin{vmatrix} 4 & 1 \\ 0 & -1 \end{vmatrix}$$

Now After sol the Answer will come

$$\boxed{-59} \text{ Ans}$$